

Climate Shocks and the Financial Health of American Households: Incidence, Persistence, and Inequality

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Abstract

Climate shocks impose financial costs on households long after the weather returns to normal, yet these costs are rarely measured in dollars, traced over time, or mapped onto the communities least able to absorb them. Using a U.S. county and state panel (2011–2023) that links NOAA climate data, PRISM humidity, EPA air quality, and CDC vulnerability indices to insurance premiums, medical debt, hospital bad debt, and local income and employment, I estimate the household financial consequences of extreme cold, heat, and drought along three dimensions: incidence, persistence, and inequality.

For incidence, two-way fixed-effects models with distributed lags, a natural-experiment difference-in-differences against never-exposed counties, and local projections show that cold and drought raise medical debt and insurance premiums one to two years out, operating through a household income channel rather than systemic health spending. The 2012 Midwest drought, used as a sharp natural experiment, lowered per-capita income by \$1,311 and employment by roughly 2,000 jobs in treated relative to never-exposed counties, identifying the income channel through which drought later surfaces as debt; random-effects specifications are rejected throughout.

For persistence, I decompose each shock into onset, persistence, and exit transitions—testing whether the cost of entering a shock is mirrored by the benefit of leaving it—and estimate cumulative-dose responses in the number of shock-years a county has accumulated. The dynamics are shock-specific: drought debt scars and does not unwind on exit, cold employment losses compound (the tenth cumulative cold-year costs roughly 5,700 more jobs than the first, echoed by a staggered difference-in-differences), and heat saturates into a standing level difference. These findings survive adjustment for humidity, are not mediated by migration or population aging, and hold under stricter extreme-temperature thresholds.

For inequality, interacting shocks with the CDC Social Vulnerability Index shows the harm to income, employment, and premiums is amplified in structurally vulnerable counties—extreme heat costs jobs, cold’s income loss is several times larger, and drought raises premiums—while credit-bureau medical debt proves a measurement-fragile outcome. The results argue that environmental risk assessment in health and insurance markets must price not only the lagged incidence of climate stress, but its persistence and its unequal distribution.

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